

Information Assurance Security

Digital Forensic: Email Artifacts

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Overview:

This report presents a simulated email forensic analysis in the absence of Outlook **.pst** or **.ost** files. Instead, artifacts from webmail usage (Gmail) were located using browser history, cookies, and downloads via FTK Imager and DB Browser. The goal was to detect phishing indicators, suspicious attachments, and trace user behavior.

Tools Used:

- **FTK Imager** – Extracted Chrome browsing history and cookies
- **DB Browser for SQLite** – Parsed Chrome's **History** and **Cookies** files
- **Google Chrome Forensics Paths:**
 - C:\Users\%userprofile%\AppData\Local\Google\Chrome\User Data\Default\History
 - C:\Users\%userprofile%\AppData\Local\Google\Chrome\User Data\Default\Cookies

Screenshot 1 – Gmail Visit History Table

DB Browser for SQLite - C:\Users\billy\AppData\Local\Google\Chrome\User Data\Default\History

File Edit View Tools Help

New Database Open Database Write Changes Revert Changes Undo Open Project Save Project Attach Database Close Database

Database Structure Browse Data Edit Pragmas Execute SQL

Table: urls mail

	id	url	title	visit_count	typed_count	last_visit_time	hidden
	Filter	Filter	Filter	Filter	Filter	Filter	Filter
1	13232	https://accounts.google.com.ph/...	Inbox (1,366) - 23100803@usc.edu.ph ...	1	0	13394517441608568	0
2	13239	https://accounts.google.com/...	Gmail	1	0	13394517393889454	0
3	13245	https://accounts.google.com/...	Inbox (749) - billyxd669@gmail.com - ...	1	0	13394517332347974	0
4	15039	https://accounts.google.com/...	Inbox (202) - dajao.j@ciccebu.edu.ph...	1	0	13395595653850857	0
5	14027	https://accounts.google.com/o/oauth2...	Mag-sign in - Google Accounts	1	0	13395053803404488	0

This screenshot shows Chrome browser history entries pointing to various Gmail-related URLs such as <https://accounts.google.com> and <https://mail.google.com>, including inbox titles and email addresses. The evidence demonstrates that multiple Gmail accounts (e.g., 23100803@usc.edu.ph, billyxd669@gmail.com, dajac.j@ciccebu.edu.ph) were accessed, confirming user email activity through webmail and account switching.

Screenshot 2: Chrome Cookie Table (Gmail Sessions)

Table: cookies																
	creation_utc	host_key	name	ited	path	expires_utc	is_secure	is_httponly	last_access_utc	has_expires	is_persistent	priority	samesite	source_scheme	source	source
	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter
1	13384451851657801	mail.google.com	__GMAIL_LF	B..	/mail/u/3	13415987851000000	1	0	13386820180399754	1	1	1	-1	2		
2	13386960385984840	mail.google.com	__GMAIL_LF	B..	/chat/u/2	13418496385000000	1	0	13386960385984840	1	1	1	-1	2		
3	13384453153823951	mail.google.com	__GMAIL_LF	B..	/chat/u/3	13415989153000000	1	0	13387014706612543	1	1	1	-1	2		
4	13384453115063467	mail.google.com	__GMAIL_LF	B..	/chat/u/0	13415989115000000	1	0	13390323799190098	1	1	1	-1	2		
5	13386819945519010	mail.google.com	__GMAIL_LF	B..	/mail/u/1	13418355945000000	1	0	13394647999353443	1	1	1	-1	2		
6	13395945985667270	mail.google.com	COMPASS	B..	/chat/u/1	13396809985667270	1	1	13395946111206878	1	1	1	0	2		
7	13394641371448951	mail.google.com	OSID	B..	/	13429201371448951	1	1	13395946197963923	1	1	1	-1	2		
8	13394641371448973	mail.google.com	__Secure-OSID	B..	/	13429201371448973	1	1	13395946197963923	1	1	1	0	2		
9	13395417733786561	mail.google.com	__Host-GMAIL_SCH_GML	B..	/	13398009733786561	1	1	13395946197963923	1	1	1	1	2		
10	13395417733786443	mail.google.com	__Host-GMAIL_SCH_GMN	B..	/	13398009733786443	1	1	13395946197963923	1	1	1	-1	2		
11	13395417733786535	mail.google.com	__Host-GMAIL_SCH_GMS	B..	/	13398009733786535	1	1	13395946197963923	1	1	1	2	2		
12	13395945986257946	mail.google.com	COMPASS	B..	/	13396809986257946	1	1	13395946197963923	1	1	1	0	2		
13	13387251570023064	mail.google.com	__GMAIL_LF	B..	/chat/u/1	13418787570000000	1	0	13395946668582585	1	1	1	-1	2		
14	13395417780076349	mail.google.com	OTZ	B..	/	13398009780000000	1	0	13395946668582585	1	1	1	-1	2		

This screenshot presents extracted cookie data from Chrome's **Cookies** SQLite database, showing session cookies tied to mail.google.com. Key cookies such as **__GMAIL_LF**, **COMPASS**, and other secure flags suggest multiple active Gmail sessions across different tabs or users. These cookies also reflect account activity, chat usage, and user authentication status.

Screenshot 3: Gmail Compose and Payment Redirection History

DB Browser for SQLite - C:\Users\bill\OneDrive\Desktop\ias\History							
File Edit View Tools Help							
New Database Open Database Write Changes Revert Changes Undo Open Project Save Project Attach Database Close Database							
Database Structure Browse Data Edit Pragmas Execute SQL							
Table: urls mail							
	id	url	title	visit_count	typed_count	last_visit_time	hidden
	Filter	Filter	Filter	Filter	Filter	Filter	Filter
1	11458	https://mail.google.com/mail/?...	Compose Mail - 23100803@usc.edu.ph -...	1	0	13392991483432700	0
2	11459	https://mail.google.com/mail/u/1/?...	Compose Mail - 23100803@usc.edu.ph -...	1	0	13392991483432700	0
3	11460	https://mail.google.com/mail/u/1/?...	Compose Mail - 23100803@usc.edu.ph -...	1	0	13392991483432700	0
4	11461	https://mail.google.com/mail/u/1/?...	Compose Mail - 23100803@usc.edu.ph -...	1	0	13392991483432700	0
5	11462	https://mail.google.com/mail/u/1/?...	Compose Mail - 23100803@usc.edu.ph -...	1	0	13392991483432700	0
6	11485	https://mail.google.com/mail/u/1/?...	Mental Health Awareness Campaign - ...	1	0	13392994717888969	0
7	11486	https://mail.google.com/mail/u/1/?...	Mental Health Awareness Campaign - ...	1	0	13392994717888969	0
8	11489	https://mail.google.com/mail/u/1/...	Mental Health Awareness Campaign - ...	1	0	13392994756426053	0
9	11488	https://mail.google.com/mail/u/1/...	Mental Health Awareness Campaign - ...	2	0	13392994800831863	0
10	11912	https://gw.dragonpay.ph/Pay.aspx?...	Dragonpay Online Payment System	1	0	13393665275223575	0
11	11916	https://gw.dragonpay.ph/Bank/...	Dragonpay - Offline Banking Payment	1	0	13393666898877689	0
12	11921	https://mail.google.com/mail/u/2/...	Payment Instruction for Transaction ...	1	0	13393666934306865	0
13	11922	https://www.google.com/url?q=https://...	Dragonpay Payment Instruction	1	0	13393666941803887	0
14	11923	https://gw.dragonpay.ph/Bank/...	Dragonpay Payment Instruction	1	0	13393666941803887	0
15	12124	https://login.microsoftonline.com/...	Continue	1	0	13394028279793962	0
16	12125	https://login.live.com/...	Continue	1	0	13394028279793962	0
17	12265	https://www.youtube.com/watch?...	(148) Herald Clockwork eats 19 ...	1	0	13394189867431964	0
18	12267	https://www.youtube.com/watch?...	(148) Herald Clockwork eats 19 ...	1	0	13394190974092632	0
19	13245	https://accounts.google.com/...	Inbox (749) - billyxd669@gmail.com -...	1	0	13394517332347974	0
20	13246	https://mail.google.com/accounts/...	Inbox (749) - billyxd669@gmail.com -...	1	0	13394517332347974	0
21	13247	https://mail.google.com/mail/u/0/?...	Inbox (749) - billyxd669@gmail.com -...	1	0	13394517332347974	0
22	13239	https://accounts.google.com/...	Gmail	1	0	13394517393889454	0
23	13238	https://accounts.google.com/v3/...	Gmail	2	0	13394517395572587	0
24	13237	https://accounts.google.com/v3/...	Gmail	1	0	13394517402665782	0
25	13236	https://accounts.google.com/v3/...	Gmail	1	0	13394517422041251	0
26	13230	https://mail.google.com/accounts/...	Inbox (1,366) - 23100803@usc.edu.ph ...	1	0	13394517441608568	0
27	13231	https://accounts.youtube.com/...	Inbox (1,366) - 23100803@usc.edu.ph ...	1	0	13394517441608568	0
28	13232	https://accounts.google.com.ph/...	Inbox (1,366) - 23100803@usc.edu.ph ...	1	0	13394517441608568	0
29	13225	https://accounts.google.com/o/oauth2...	Mag-sign in - Google Accounts	1	0	13394517502429548	0
30	13223	https://accounts.google.com/o/oauth2...	Mag-sign in - Google Accounts	1	0	13394517504363950	0
31	13222	https://accounts.google.com/o/oauth2...	Mag-sign in - Google Accounts	1	0	13394517504375145	0

This browser history snapshot displays visits to Gmail's Compose interface (<https://mail.google.com/mail/u/1/>) for the account 23100803@usc.edu.ph, along with navigation to payment-related sites such as [dragonpay.ph](https://gw.dragonpay.ph). This sequence suggests that the user may have engaged in composing messages and potentially followed links to external financial platforms — a behavior typical in phishing or scam scenarios.

Date	Sender (Account Used)	Recipient / Domain Accessed	Subject / Page Title	Suspicious ?	Notes
Jun 27, 2025	23100803@usc.edu.ph	mail.google.com	Compose Mail	No	Normal compose activity for school email
Jun 27, 2025	23100805@usc.edu.ph	mail.google.com	Compose Mail	No	Another student account; shows Gmail was actively used
Jun 27, 2025	billyxd669@gmail.com	mail.google.com/accounts	Inbox (749)	Maybe	Personal Gmail accessed on same machine; used to receive messages
Jun 27, 2025	Unknown	dragonpay.ph	Payment Instruction	Yes	Transaction initiated via Gmail session – potential scam/phishing
Jun 27, 2025	Unknown	mail.google.com	Mental Health Awareness Campaign	Maybe	Could contain attachments or links – follow-up needed

Timeline of Events (Based on Timestamps)

Time (UTC/forensic)	Event
13392991483432700 May 29, 2025 – 11:24:43 AM	Gmail Compose opened with usc.edu.ph account
13393666934306865 June 6, 2025 – 07:02:14 AM	Dragonpay payment instructions viewed
13394517332347974 June 16, 2025 – 03:15:32 AM	billyxd669@gmail.com inbox accessed
13394517441608568 June 16, 2025 – 03:17:21 AM	23100803@usc.edu.ph inbox accessed (repeated logins)
13395053803404488 June 22, 2025 – 08:16:43 AM	OAuth sign-in to Gmail or Google service

Conclusion & Recommendations:

The browsing history and cookie artifacts confirm the use of **multiple Gmail accounts**, some of which were used to compose and read emails within the same session. The presence of financial transaction sites like Dragonpay immediately after Gmail sessions raises red flags, suggesting the user may have clicked on payment links from emails, possibly phishing attempts.

Recommendations:

- Perform dynamic malware scan on downloaded ZIP or invoice files (if any)
- Monitor webmail usage on secured systems

Proof of the living

